Chapter 15 : Body Fluids and Circulation

1. Cardiac activities of the heart are regulated by: **[NEET-2025]**

   A. Nodal tissue

   B. A special neural centre in the medulla oblongata

   C. Adrenal medullary hormones

   D. Adrenal cortical hormones

   Choose the correct answer from the options given below :

   (1) A, B and C Only

   (2) A, B, C and D

   (3) A, C and D Only

   (4) A, B and D Only

2. A person with blood group ARh– can receive the blood transfusion from which of the following types?

   A. BRh– **[Re-NEET-2024]**

   B. ABRh–

   C. ORh–

   D. ARh–

   E. ARh+

   Choose the correct answer from the options given below :

   (1) D and E only

   (2) D only

   (3) A and B only

   (4) C and D only

3. 'Lub' sound of Heart is caused by the___________. **[Re-NEET-2024]**

   (1) closure of the semilunar values

   (2) opening of tricuspid and bicuspid values

   (3) opening of the semilunar valves

   (4) closure of the tricuspid and bicuspid valves

4. The mother has A+ blood group the father has B+ and the child is A+. What can be the possibility of genotypes of all three, respectively? **[Re-NEET-2024]**

A. IAIA | IBi | IBi
B. IAIA | IBi | IAi
C. IBi | IAIA | IAIB
D. IAIA | IBIB | IAi
E. IAi | IBi | IAi

Choose the correct answer from the option given below:

(1) C and D
(2) D and A
(3) A and B
(4) B and E

5. Following are the stages of pathway for conduction of an action potential through the heart **[NEET-2024]**

A. AV bundle
B. Purkinje fibres
C. AV node
D. Bundle branches
E. SA node

Choose the correct sequence of pathway from the options given below

(1) E-C-A-D-B
(2) A-E-C-B-D
(3) B-D-E-C-A
(4) E-A-D-B-C

6. Match List I with List II : **[NEET-2024]**

List I	List II
A. P wave	I. Heart muscles are electrically silent.
B. QRS complex	II. Depolarisation of ventricles.
C. T wave	III. Depolarisation of atria.
D. T-P gap	IV. Repolarisation of ventricles.

Choose the correct answer from the options given below :

(1) A-I, B-III, C-IV, D-II
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-IV, B-II, C-I, D-III

7. Given below are two statements : Statement I : **[NEET-2024]**

Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.

Statement II : Both bone marrow and thymus provide micro environments for the development and maturation of T-lymphocytes.

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

8. As per ABO blood grouping system, the blood group of father is B+, mother is A+ and child is O+. Their respective genotype can be **[NEET-2024]**

A. IBi/IAi/ii

B. IBIB/IAIA/ii

C. IAIB/iIA/Ibi

D. IAi/IBi/IAi

E. iIB/iIA/IAIB

Choose the most appropriate answer from the options given below :

(1) A only

(2) B only

(3) C & B only

(4) D & E only

9. Match List-I with List-II **[NEET-2023-MANIPUR]**

List-I (ECG)	List-II (Electrical activity of heart)
(A) P-wave	(I) Depolarisation of ventricles
(B) QRS complex	(II) End of systole
(C) T wave	(III) Depolarisation of atria
(D) End of T wave	(IV) Repolarisation of ventricles

Choose the correct answer from the options given below :

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(4) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

10. Match List - I with List - II. **[NEET-2023-MANIPUR]**

List – I	List - II
(A) Eosinophils	(I) 6 - 8%
(B) Lymphocytes	(II) 2 - 3%
(C) Neutrophils	(III) 20 - 25%
(D) Monocytes	(IV) 60 - 65%

Choose the correct answer from the options given below :

(1) A-(II), B-(III), C-(IV), D-(I)

(2) A-(II), B-(III), C-(I), D-(IV)

(3) A-(IV), B-(I), C-(II), D-(III)

(4) A-(IV), B-(I), C-(III), D-(II)

11. Match List I with List II. **[NEET-2023]**

List I	List II
A. P-wave	I. Beginning of systole
B. Q-wave	II. Repolarisation of ventricles
C. QRS complex	III. Depolarisation of atria
D. T-wave	IV. Depolarisation of ventricles

Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV, D-II   (2) A-IV, B-III, C-II, D-I

(3) A-II, B-IV, C-I, D-III   (4) A-I, B-II, C-III, D-IV

12. Which of the following statements are correct? **[NEET-2023]**

A. Basophils are most abundant cells of the total WBCS

B. Basophils secrete histamine, serotonin and heparin

C. Basophils are involved in inflammatory response

D. Basophils have kidney shaped nucleus

E. Basophils are agranulocytes

Choose the correct answer from the options given below:

(1) D and E only

(2) C and E only

(3) B and C only

(4) A and B only

13. A unique vascular connection between the digestive tract and liver is called ________ . **[Re-NEET-2022]**

(1) Hepato–pancreatic system

(2) Hepatic portal system

(3) Renal portal system

(4) Hepato–cystic system

14. Arrange the following formed elements in the decreasing order of their abundance in blood in humans :

(a) Platelets **[Re-NEET-2022]**

(b) Neutrophils

(c) Erythrocytes

(d) Eosinophils

(e) Monocytes

Choose the most appropriate answer from the options given below :

(1) (c), (a), (b), (e), (d)   (2) (c), (b), (a), (e), (d)

(3) (d), (e), (b), (a), (c)   (4) (a), (c), (b), (d), (e)

15. Which one of the following statements is correct ? **[NEET-2022]**

(1) The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria

(2) Blood moves freely from atrium to the ventricle during joint diastole.

(3) Increased ventricular pressure causes closing of the semilunar valves.

(4) The atrio-ventricular node (AVN) generates an action potential to stimulate atrial contraction

16. Given below are two statements: **[NEET-2022]**

Statement I: The coagulum is formed of network of threads called thrombins.

Statement II: Spleen is the graveyard of erythrocytes.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

17. Persons with 'AB' blood group are called as "Universal recipients". This is due to : **[NEET-2021]**

(1) Absence of antigens A and B on the surface of RBCs

(2) Absence of antigens A and B in plasma

(3) Presence of antibodies, anti-A and anti-B, on RBCs

(4) Absence of antibodies, anti-A and anti-B, in plasma

18. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins? **[NEET-2021]**

(1) Thrombin

(2) Renin

(3) Epinephrine

(4) Thrombokinase

19. Match the following columns and select the correct option. **[NEET-2020]**

Column - I	Column - II
(a) Eosinophils	(i) Immune response
(b) Basophils	(ii) Phagocytosis
(c) Neutrophils	(iii)Release histaminase, destructive enzymes
(d) Lymphocytes	(iv) Release granules containing histamine

	(a)	(b)	(c)	(d)
(1)	(ii)	(i)	(iii)	(iv)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(i)	(ii)	(iv)	(iii)

20. Which of the following conditions cause erythroblastosis foetalis ? **[NEET-2020 COVID-19]**

(1) Mother Rh+ve and foetus Rh–ve

(2) Mother Rh–ve and foetus Rh+ve

(3) Both mother and foetus Rh–ve

(4) Both mother and foetus Rh+ve

21. Match the Column - I with Column -II **[NEET-2019]**

Column - I	Column - II
(a) P-wave	(i) Depolarisation of ventricles
(b) QRS complex	(ii) Repolarisation of ventricles
(c) T-wave	(iii) Coronary ischemia
(d) Reduction	(iv) Depolarisation of in the size atria of T-wave
	(v) Repolarisation of atria

Select the correct option –

	(a)	(b)	(c)	(d)
(1)	(iv)	(i)	(ii)	(iii)
(2)	(iv)	(i)	(ii)	(v)
(3)	(ii)	(i)	(v)	(iii)
(4)	(ii)	(iii)	(v)	(iv)

22. All the components of the nodal tissue are auto excitable. Why does the SA node act as the normal pacemakar? **[NEET-2019-ODISHA]**

(1) SA node has the lowest rate of depolarisation.

(2) SA node is the only component to generate the threshold potential.

(3) Only SA node can convey the action potential to the other components.

(4) SA node has the highest rate of depolarisation.

23. A specialised nodal tissue embedded in the lower corner of the right atrium, close to Atrio-ventricular septum, delays the spreading of impulses to heart apex for about 0.1 sec. The delay allows.

(1) blood to enter aorta. **[NEET-2019-ODISHA]**

(2) the ventricles to empty completely.

(3) blood to enter pulmonary arteries.

(4) the atria to empty completely.

24. Match the items given in Column I with those in Column II and select the correct option given below:

Column I	Column II
a. Tricuspid valve	i. Between left atrium and left ventricle
b. Bicuspid valve	ii. Between right ventricle and pulmonary artery
c. Semilunar valve	iii. Between right atrium and right ventricle

[NEET-2018]

	a	b	c
(1)	iii	I	ii
(2)	I	iii	ii
(3)	i	ii	iii
(4)	ii	i	iii

25. Match the items given in Column I with those in Column II and select the correct option given below:-

Column I	Column II
a. Fibrinogen	i. Osmotic balance
b. Globulin	ii. Blood clotting
c. Albumin	iii. Defence mechanism

[NEET-2018]

	a	b	c
(1)	iii	ii	i
(2)	i	ii	iii
(3)	i	iii	ii
(4)	ii	iii	i

26. Adult human RBCs are enucleated. Which of the following statement(s) is/are most appropriate explanation for this feature ? **[NEET-2017]**

(a) They do not need to reproduce

(b) They are somatic cells

(c) They do not metabolize

(d) All their internal space is available for oxygen transport

(1) only (a)
(2) (a), (c) and (d)
(3) (b) and (c)
(4) only (d)

27. The hepatic portal vein drains blood to liver from : **[NEET-2017]**

(1) Stomach

(2) Kidneys

(3) Intestine

(4) Heart

28. Frog's heart when taken out of the body continues to beat for sometime. Select the best option from the following statements. **[NEET-2017]**

(a) Frog is a poikilotherm.

(b) Frog does not have any coronary circulation.

(c) Heart is "myogenic" in nature.

(d) Heart is autoexcitable

Options:

(1) Only(d)

(2) (a) and (b)

(3) (c)and(d)

(4) Only(c)

29. Blood pressure in the pulmonary artery is :- **[NEET- I 2016]**

(1) same as that in the aorta.

(2) more than that in the carotid.

(3) more than that in the pulmonary vein.

(4) less than that in the venae cavae.

30. Name the blood cells, whose reduction in number can cause clotting disorder, leading to excessive loss of blood from the body. **[NEET- II 2016]**

(1) Neutrophils

(2) Thrombocytes

(3) Erythrocytes

(4) Leucocytes

31. Serum differs from blood in :- **[NEET- II 2016]**

(1) Lacking clotting factors

(2) Lacking antibodies

(3) Lacking globulins

(4) Lacking albumins

ANSWER KEY

1. (1)
2. (4)
3. (4)
4. (4)
5. (1)
6. (2)
7. (1)
8. (1)
9. (2)
10. (1)
11. (1)
12. (3)
13. (2)
14. (1)
15. (2)
16. (3)
17. (4)
18. (1)
19. (2)
20. (2)
21. (1)
22. (4)
23. (4)
24. (1)
25. (4)
26. (4)
27. (3)
28. (3)
29. (3)
30. (2)
31. (1)